

David Feldman

CA → Urbana, IL | dodya.feldman@gmail.com | github.com/dodyya | linkedin.com/in/dodyya/ | dodyya.github.io/

EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

B.S. Mathematics and Computer Science - GPA: 4.0

Expected Grad May 2028

- State Farm CS Scholarship: 1 of 2 from a cohort of 2,500+ recognized for exceptional academic promise
- Societies: James Scholar Honors Program, Quant @ Illinois (Software Division), SIAM, ACM SIGma, SIGPLAN
- Coursework: Computer Architecture, Computational Geometry, Algorithms, Linear Algebra, Probability, Data Structures

RESEARCH

Illinois Mathematics Lab

Jan 2026 — May 2026

UIUC Math Department

Urbana, IL

- Built optimized Rust and Python tools for homotopy groups of spheres that computed further into the EHP spectral sequence than the algorithm's original author
- Produced the most detailed visualizations of the sequence to date; presented findings in a research poster ([repo](#))

SELECTED PROJECTS

Real-Time Audio Visualization Suite

Nov 2025

- Manual Rust implementation of iterative Cooley-Tukey FFT; real-time spectrogram with synchronized WAV playback
- Lock-free SPSC ring buffer for cross-thread audio streaming; memory-mapped WAV parser with zero-copy sample reinterpretation and lazy bidirectional sliding-window cache

High Performance Particle Physics Engine

Jun 2026

- Multithreaded Rust physics engine simulating kinematics, gravity, and collisions for 50,000+ particles at 60 FPS
- Lock-free atomics, grid spatial partitioning and sweep, Verlet integration; ported to CUDA kernels (cuda-oxide) with CUDA-OpenGL interop keeping particle state GPU-resident with no CPU round-trip

Eulerian Fluid Simulation

Aug 2025

- Interactive real-time 2D CFD simulator with incompressible flow, obstacle interaction, and multiple visualization modes
- MAC grid velocity field with backward advection sampling, divergence-free projection for pressure solving in Rust

Interactive Piano-Mover's Problem Visualization

May 2026

- C++, CGAL motion planner for a translating and rotating 2D polygonal robot: live editor alongside 3D path in free space
- Construction of (x, y, θ) configuration space by Minkowski sum of robot with obstacles; tetrahedral subdivision, Dijkstra

Automatic Differentiation Engine

Jul 2025

- Reverse-mode automatic differentiation engine in Rust; topological DAG sort for the gradient of an arbitrary expression
- Demonstrated with end-to-end neural network training via backpropagation

Constraint Satisfaction Puzzle Solver

Jul 2025

- [Flow Free](#) game clone and solver in Rust, backtracking and forced move greedy heuristics, Python puzzle collection pipeline

End-to-end Mechanical Keyboard

Dec 2023

- Fully custom printed PCB with QMK/C, community group buy and open-sourced case hardware and firmware

SKILLS

- **Languages:** Rust, C/C++, CUDA, MIPS Assembly, Python, OCaml, Typst, LaTeX
- **Tools:** Git, Cargo, rayon, cargo-flamegraph, PyTorch, NumPy, SciPy, pandas, Matplotlib
- **Concepts:** Optimization, Automatic Differentiation, GPU Programming, Linear Algebra, Probability

EXPERIENCE

Full-Stack Developer

Sep 2023 — May 2025

Los Altos Hacks

Los Altos, CA

- Built and maintained software infrastructure for the world's largest high school hackathon, two years running
- Led the judging-system sub-team, implementing custom pairwise voting to streamline in-hackathon scoring